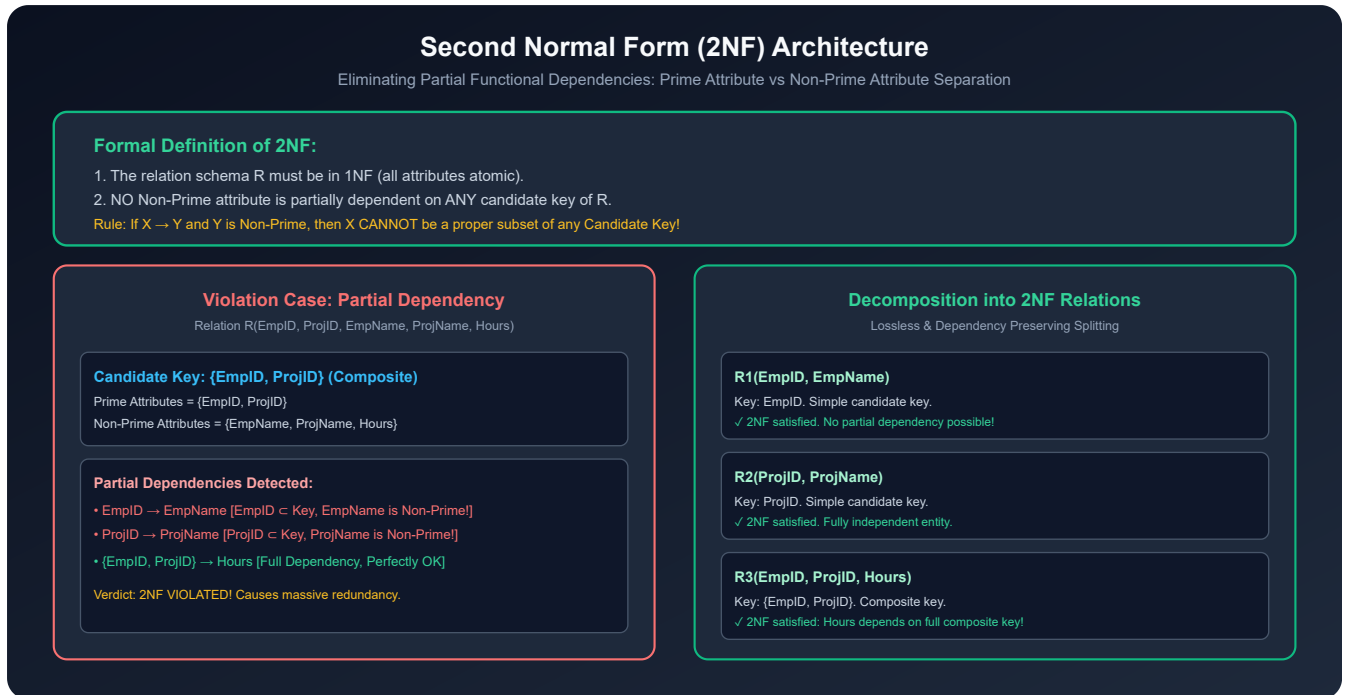


Module 06: Second Normal Form (2NF)



> **Course:** Database Management Systems (AKTU Code: **BCS501**)

****Topic Focus:**** Second Normal Form Formal Criteria, Full Functional Dependency, Partial Functional Dependency Violation, Special Theorem for Simple Keys, 2NF Decomposition Algorithm with AKTU Solved Problem.

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1. Second Normal Form (2NF) Formal Definition

Ek relation schema R **Second Normal Form (2NF)** me tab hota hai jab:

1. Wo relation already **1NF** me ho.
2. Us relation me **koi bhi Partial Functional Dependency** na ho.

1.1 Mathematical Formulation

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$$\rightarrow Y \in F \quad \text{such that } Y \text{ is a Non-Prime Attribute}$$

$\mathbf{X}$ ext{ is NOT a proper subset of ANY Candidate Key of } R.

Prime Attribute: Jo kisi bhi ek candidate key ka member ho.

Non-Prime Attribute: Jo kisi bhi candidate key ka member na ho.

Partial Dependency: $\$ \text{ ext}\{\text{Proper Subset of Candidate Key}\}$

 $\rightarrow \text{ext}\{\text{Non-Prime Attribute}\}.$

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2. The Powerful "Simple Candidate Key" Theorem (AKTU Shortcut)

Theorem: Agar kisi relation ke sabhi candidate keys **Simple (Single Attribute)** hain (yaani koi bhi candidate key composite nahi hai), toh wo relation **by default hamesha 2NF me hoga**!

Proof:

Partial dependency tabhi ho sakti hai jab candidate key ka koi proper non-empty subset ho. Agar Candidate Key $K = \{A\}$ hai, toh iska koi non-empty proper subset exist hi nahi karta! Isliye partial dependency create hona mathematically impossible hai!

3. Systematic 2NF Decomposition Algorithm

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Input: Relation R with FD set F and Candidate Keys CK

Output: Set of relations in 2NF

1. Find all Candidate Keys of R.
2. Identify all Prime Attributes and Non-Prime Attributes.
3. Check every FD $X \rightarrow Y$ in F:

- If Y is Non-Prime AND X is a proper subset of some Candidate Key:

→ Partial Dependency Detected!

4. Decompose R:

- Create new relation $R_1(X, Y)$ with Primary Key X.
- Remove Y from original relation: $R_{\text{remaining}} = R - Y$.

5. Repeat until no partial dependencies remain.

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4. Architectural Diagram

5. Gateway Classes Solved AKTU Examination Problem

Problem: Given relation $R(\text{Emp_ID}, \text{Project_ID}, \text{Emp_Name}, \text{Project_Name}, \text{Hours})$ with functional dependencies:

$F = \{ \text{Emp_ID} \rightarrow \text{Project_ID}, \text{Project_ID} \rightarrow \text{Hours} \}$

$\text{Emp_ID} \rightarrow \text{Hours}, \text{Emp_ID} \rightarrow \text{Project_ID}$

$\text{Emp_Name} \rightarrow \text{Project_ID}$

$\rightarrow \text{Project_Name} \rightarrow \text{Emp_ID}$

1. Find Candidate Key:

- $\{ \text{Emp_ID}, \text{Project_ID} \}^+ = \{ \text{Emp_ID}, \text{Project_ID}, \text{Hours}, \text{Emp_Name}, \text{Project_Name} \} = R$.

- Candidate Key = $\{ \text{Emp_ID}, \text{Project_ID} \}$.

2. Classify Attributes:

- **Prime Attributes:** $\{ \text{Emp_ID}, \text{Project_ID} \}$

- **Non-Prime Attributes:** $\{ \text{Emp_Name}, \text{Project_Name}, \text{Hours} \}$

3. Test Each Functional Dependency for 2NF:

- $\{ \text{Emp_ID}, \text{Project_ID} \}$

$\rightarrow \text{Hours}$: Determinant is Full Candidate Key. **(Full FD - OK)**

- Emp_ID

$\rightarrow \text{Emp_Name}$: LHS $\text{Emp_ID} \subset \text{Candidate Key}$ and RHS Emp_Name is Non-Prime! **(Partial FD - VIOLATION!)**

- Project_ID

$\rightarrow \text{Project_Name}$: LHS $\text{Project_ID} \subset \text{Candidate Key}$ and RHS Project_Name is Non-Prime! **(Partial FD - VIOLATION!)**

4. Decomposition into 2NF:

- $R_1(\text{Emp_ID}, \text{Emp_Name})$ with Key = Emp_ID .

- $R_2(\text{Project_ID}, \text{Project_Name})$ with Key = Project_ID .

- $R_3(\text{Emp_ID}, \text{Project_ID}, \text{Hours})$ with Key = $\{ \text{Emp_ID}, \text{Project_ID} \}$.

5. **Conclusion:** All three relations R_1, R_2, R_3 are strictly in 2NF, lossless, and dependency preserving!